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A New Late Cretaceous Paleomagnetic Pole from the Adel Mountains, West Central Montana

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North America's apparent polar wander path has been poorly defined between the mid-Cretaceous and Paleocene reference pole positions. Existing data allowed 13° of apparent polar motion over about 22 m.y. (87–65 Ma) roughly coinciding with the beginning of Laramide deformation (~80 Ma). We report on a paleomagnetic study of the Adel Mountain Volcanic rocks to refine the North American apparent polar wander path for this interval. The shonkinite rocks of the Adel Mountain Volcanic field are on the eastern edge of the Cretaceous-Paleocene fold and thrust belt; some of these structures disturb the western edge of the volcanic pile. We obtained two new K-Ar dates from the Adel rocks. One date, on biotite (from a shonkinite dike that crosscuts most of the volcanic rocks, is 71.2 ± 2.7 Ma. The other, a whole rock date from a flow deep in the volcanic pile, is 81.1 ± 3.5 Ma. We collected six to nine paleomagnetic samples from each of 34 sites in roadcuts and natural outcrops of flows, dikes, and laccoliths. Positive fold and conglomerate tests, along with alternating field and thermal demagnetization, indicate that our characteristic remanent directions are primary magnetizations acquired before Late Cretaceous to Paleocene thrust belt deformation. Averaging the virtual geomagnetic poles from 26 reliable sites, all of normal polarity, yields a paleopole at 82.2°N, 209.9°E ($\alpha 95 = 6.80°$, $k = 18.38$). This pole is concordant with the Paleocene reference pole (82.0°N, 170.2°E, $\alpha 95 = 3.5°$, $k = 18.6$ (Diehl et al., 1983)) and is 11.6° from the Globerman and Irving (1988) mid-Cretaceous pole at 71°N, 196°E. The youngest information in the Cretaceous stillstand pole is from the Niobrara Formation (Shive and Frerichs, 1974) at about 85–89 Ma. If we take the average age of the Adel Mountain Volcanics to be 76 Ma, then ~12° of apparent polar motion occurred between 87 Ma and 76 Ma. Thus, rapid apparent polar motion correlates well with the onset of Laramide deformation.

INTRODUCTION

Paleomagnetic pole positions and apparent polar wander (APW) paths for stable cratons provide an unequaled reference frame for delineating large-scale relative motions and tectonic histories of crustal fragments. This reference frame requires precise APW paths assembled from a series of accurate reference poles with well known ages. Although North American reference poles for the mid-Cretaceous and Early Tertiary are well established, the Late Cretaceous segment of the APW path for North America is poorly defined.

Cusps in APW paths separate smooth tracks representing uniform plate motion. Cusps are manifestations of major plate interactions [*Gordon et al.*, 1984; *May and Butler*, 1986] which may be correlated with specific orogenic events [*Irving and Park*, 1972; *Coney*, 1972, 1978]. The youngest well-defined APW track for North America, the J2-K track of *May and Butler* [1986], records rapid plate motion from the Late Jurassic to the Early Cretaceous (ca. 150–130 Ma). The J2-K track is punctuated by a mid-Cretaceous stillstand pole located at 71°N, 196°E, $\alpha 95 = 4.9°$ [*Globerman and Irving*, 1988] (Figure 1). Current data also indicate a Tertiary APW track with a different trend than the J2-K track. *Diehl et al.* [1983] provide high-quality reference pole positions for the Paleocene (67–55 Ma) (81.5°N, 192.6°E, $\alpha 95 = 3.2°$) and Eocene (54–44 Ma) (82.8°N, 170.4°E, $\alpha 95 = 3.0°$). However, these two poles along with the mean Oligocene–Miocene (40–20 Ma) pole [*Diehl et al.*, 1988] at (81.5°N, 147.3°E, $\alpha 95 = 2.4°$) have minimal angular separation (Figure 1). Because the Early Tertiary reference poles are not significantly different from each other, this APW track is not particularly well defined. Regardless, the Paleocene and Eocene paleomagnetic poles are well removed (~11°) from the mid-Cretaceous reference pole.

The cusp between the J2-K track and the Early Tertiary track/poles coincides with many interesting geologic developments in North America. These include (1) the beginning of Late Cretaceous–Eocene (Laramide) folding and thrusting about 80 Ma [*Coney*, 1972, 1978], (2) the end of major magmatism and volcanism extending from the Northern Rocky Mountains to the Sierra Nevada, (3) a decrease in the subduction angle of the Farallon Plate [*Engebretson et al.*, 1984], and (4) an increased rate of convergence between the North American and Farallon plates [*Engebretson et al.*, 1984]. Refining the APW path for this crucial interval, thereby accurately locating the cusp, is necessary for an accurate tectonic history of the western Cordillera.

The youngest poles included in the mid-Cretaceous stillstand pole [*Globerman and Irving*, 1988] are from the 88–100 Ma alkalic intrusives of Arkansas and the 85–89 Ma Niobrara Formation of Wyoming, Colorado, and Kansas [*Shive and Frerichs*, 1974]. The oldest of *Diehl et al.'s* [1983] Paleocene poles are from the 67–62 Ma Moccasin Mountains and Judith Mountains of north central Montana. Using *Globerman and Irving's* [1988] preferred mid-Cretaceous pole position and *Diehl et al.'s* [1983] Paleocene pole, about 11° of apparent motion of North America occurred over a minimum of 18 m.y. during the Late Cretaceous. This period covers the link between the J2-K track and the Tertiary poles. Alternatively, *Swenson and McWilliams* [1989] contribute two new Late Cretaceous–Paleocene poles from the Maudlow and Livingston formations of Montana and suggest that the Jurassic–Cretaceous APW path continued farther to the present-day south and east than indicated by previous workers [*Diehl et al.*, 1983; *Gordon et al.*, 1984; *May and Butler*, 1986]. To delineate between these two proposals and to better understand

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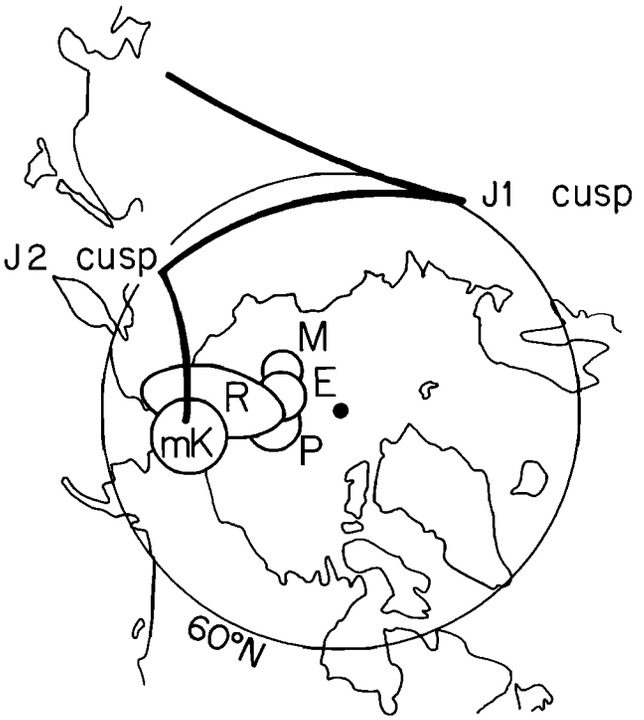


Fig. 1. Stereographic projection centered on the current spin axis (solid circle) showing paleomagnetic poles discussed in text and their associated 95% confidence limits: M, Oligocene-Miocene; E, Eocene; P, Paleocene; R, Late Cretaceous Roskruge volcanic rocks; mK, middle Cretaceous.

Late Cretaceous apparent polar wander for North America, further Late Cretaceous paleomagnetic results are required.

The Adel Mountain volcanic field is well situated to limit the gap separating the J2-K track from the Paleocene pole position. Two new K-Ar radiometric dates indicate an average age of about 76 Ma. The Adel Mountains are largely east of the Late Cretaceous to Paleocene thrusting and Eocene to recent extension which affects most of Montana west of the volcanic field. We completed a detailed paleomagnetic study of the Late Cretaceous Adel Mountain volcanic field and discuss our results below. The results indicate that the 11° of apparent motion of North America may have occurred in as little as 9 m.y.

GEOLOGY

The Adel Mountain volcanic field (latitude 47.20°N, longitude 111.85°W) covers approximately 900 km^2 (Figure 2). The volcanic rocks unconformably overlie a folded and thrusted Cretaceous sedimentary section and occupy a structural depression which parallels the eastern edge of Montana's disturbed belt. The southwestern edge of Adel Mountain volcanic field is deformed by Late Cretaceous to Paleocene folds and imbricate thrusts. The Adel Mountains were first mapped by *Lyons* [1944] as a 1000-m-thick Late Cretaceous volcanic sequence comprised primarily of large volumes of shonkinite (trachybasalt), a potassium-rich basalt of approximately equal proportions of K-feldspar and pyroxene. *Schmidt* [1972*a*,*b*, 1977, 1978], *Mudge et al.* [1982], and *Soward* [1975*a*,*b*] provide modern high-quality geologic maps of the Adel Mountain volcanic field and surrounding area. *Swenson* [1988] completed a detailed geologic map of the southern portion of the Adel Mountain volcanic field, verifying and extending the stratigraphic and structural relations of *Lyons* [1944].

The Adel Mountain volcanic rocks include flows, flow breccias, volcanic conglomerates, and associated intrusives, all of similar composition. The extrusive rocks are dominated by flow breccias and volcanic conglomerates, particularly in the southwestern portion of the field. As in many volcanic fields, flows and breccias are poorly bedded, but structural attitudes can usually be measured by examining outcrops both mesoscopically and megascopically. The entire extrusive package has been mildly deformed and mineralogically altered by zeolitic replacements and oxidation [*Lyons*, 1944; *Beall*, 1973].

The intrusive rocks, further described by *Beall* [1972, 1973] and *Whiting* [1977], include dikes, sills, plugs, other irregular bodies, and two concentric arcs of laccoliths which lie 18 and 23 km north of the volcanic center (Figure 2). The most striking feature of the entire volcanic assemblage is the enormous number of dikes which emanate radially from its center, almost all crosscutting the extrusives. These dikes range from tens of centimeters to tens of meters in thickness. The larger dikes make obvious topographic features on 1:62,500 maps for up to 30 km; they often feed tangentially attached laccoliths [*Hyndman and Alt*, 1987]. These late dikes are undeformed and are almost always vertical.

Various authors, and local geologists, have considered the Adel Volcanic rocks to be Late Cretaceous and/or Paleocene, some even assigned them to the Eocene. The Adel Volcanic rocks were first dated as Late Cretaceous based on two plant fossils found in the volcanic conglomerates [*Lyons*, 1944]. Support for a Cretaceous age was added in a March 1984 USGS Geologic Division Newsletter which credited *Mehnert and Cebula* with obtaining K-Ar dates (unpublished) of about 75 ± 2.3 Ma and 72.9 ± 2.1 Ma on hornblendes and biotites from samples of trachybasalt. The samples for these K-Ar dates were taken in 1969–1970 from *Lyons'* [1944] lower breccia unit near Wolf Creek, Montana. *Lyons* [1944] considered these volcanic rocks to be the basal portion of the Adel Mountain volcanic rocks. However, this breccia unit is now recognized as being a member of the Two Medicine Formation, which is related to the Late Cretaceous Elkhorn Mountains volcanic field [*Schmidt*, 1972*a*], not the Adel Mountain volcanic field. Thus the two radiometric dates from the USGS newsletter are probably not from rocks of the Adel Mountain volcanic field.

To add credence to the dating of the Adel Mountain volcanic rocks, we obtained two new K-Ar age dates. A whole rock date of 81.1 ± 3.5 Ma from a flow deep in the volcanic pile (site 88AV12) probably represents very early volcanism. Biotites from one of the late shonkinite dikes that crosscut the volcanic pile represents late igneous activity at 71.2 ± 2.7 Ma. These dates (see appendix for details) help constrain Adel Mountain volcanic activity to the Late Cretaceous (ca. 81–71 Ma).

FIELD AND LABORATORY PROCEDURES

All paleomagnetic measurements are expressed in SI units of induction (B, tesla) and magnetic dipole moment per unit volume (M, amperes per meter). An induction of 1T corresponds to a cgs magnetic field (H) of 10^4 Oe, while a magnetic dipole moment per unit volume of 1 A/m corresponds to a cgs magnetization (J) of 10^{-3} emu/cm^3.

Field Procedures

We collected a total of 277 samples from 34 paleomagnetic sites in roadcuts and natural outcrops in the Adel Mountain volcanic flows and intrusives. Each paleomagnetic site is presumed to represent an independent spot reading of the ancient geomagnetic field. Because of the difficulty in determining precise stratigraphic relationships in volcanic fields, we chose site locations in a roughly linear cross-section

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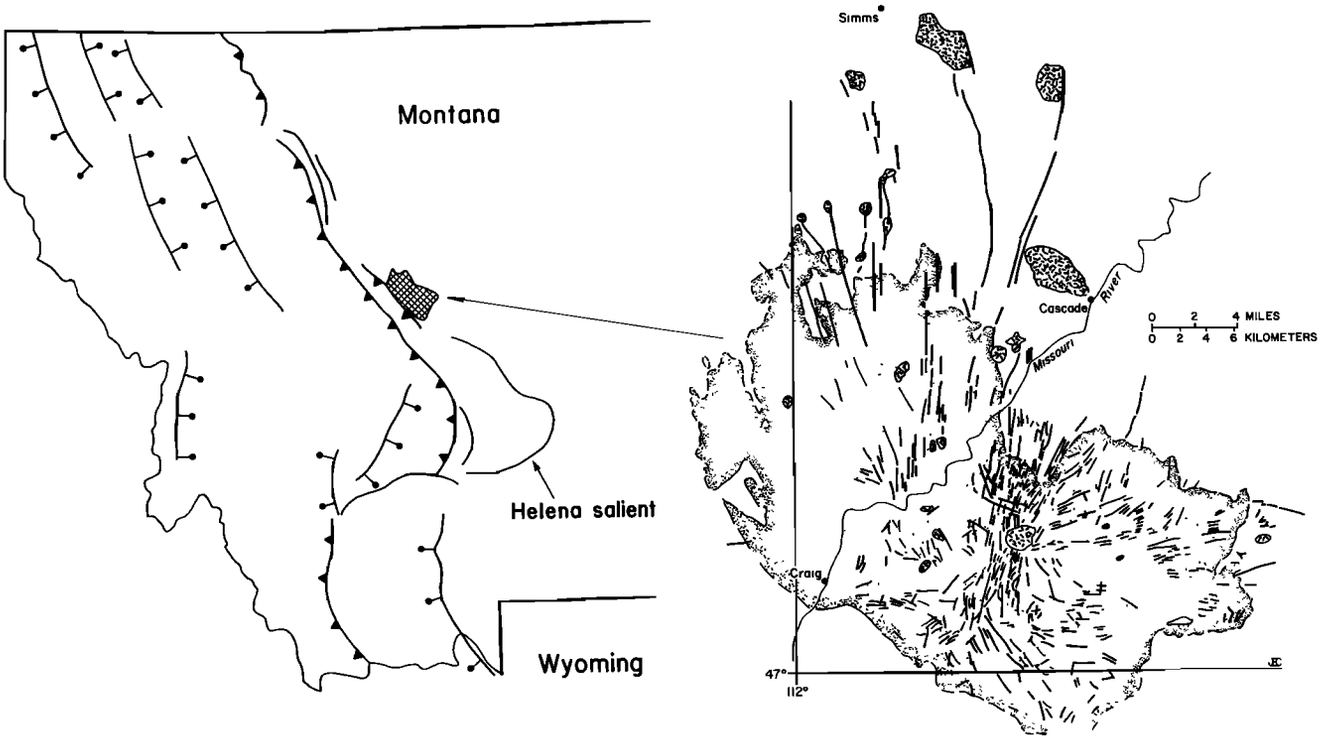


Fig. 2. Index map showing generalized tectonics of western Montana and a simplified geologic map of the Adel Mountain volcanic field [from *Hyndman and Alt*, 1987]. On the right, the stippled area is the volcanic rocks; heavy lines are dikes. The hachured areas are shonkinite intrusions; those north of the volcanic pile are laccoliths.

of the volcanic pile along the Missouri River (Figure 2). This cross-section cuts across the pre-volcanic structural depression [*Mudge et al.*, 1982] and maximizes the number of individual units and time period we sampled. At several localities, two or three sites were gathered where large dipping stacks of flows provide a number of discrete units and excellent structural control. These sites were generally separated by at least 100 m of stratigraphic thickness and three to five distinct flows to ensure sampling of independent cooling units. Nearly all (97%) of our cores were collected using a portable drill, a magnetic compass, and where possible a sun compass. However, at two sites oriented hand samples were collected from the field and cored in the lab.

Laboratory Procedures

We employed progressive alternating field (af) and/or thermal cleaning techniques to evaluate the natural remanent magnetization (NRM) of each specimen. Magnetization was measured with a Schonstedt SSM-2A spinner magnetometer. NRM intensities ranged from 0.01 to 100 A/m, well above the magnetometer's practical limit for measurement of about 10^{-4} A/m. We began by cleaning two pilot specimens from each site using separate af and thermal techniques to determine paleomagnetic stability, resolve multiple components, and learn their gross magnetic mineralogy. The remaining specimens from each site were treated to whichever method(s) best separated primary and secondary components during progressive demagnetization. Specimens were typically subjected to 10–15 steps of af cleaning from 2.5–100 mT, using a Molspin two-axis tumbling demagnetizer capable of inductions up to 100 mT. A homemade thermal demagnetizing unit was employed for thermal cleaning from 100°C to 680°C, usually in 12 or 13 steps of 15–100°C. Specimens were held at the demagnetization temperature for 25 min and cooled in a magnetically shielded space which has maximum induction less than 6 nT in the sample area. To monitor changes in magnetic mineralogy during thermal cleaning, we measured bulk susceptibilities with a Bartington M.S.2 magnetic susceptibility meter after each step.

After reducing specimen intensities to a few percent of their original magnetization, we identified characteristic remanent directions using standard orthogonal demagnetization diagrams [e.g., *Zijderveld*, 1967] and principal component analysis [*Kirschvink*, 1980]. Many samples had low coercivity, sometimes present-day, overprints. However, our cleaning techniques isolated single component characteristic remanent magnetizations for nearly every specimen. For 28 sites with single component final magnetizations, mean directions and associated statistical data were computed using the methods of *Fisher* [1953]. The other six sites were dominated by samples that did not yield reliable characteristic remanent directions after magnetic cleaning but did yield well-defined demagnetization planes. These six sites were analyzed with principal component analysis of demagnetization planes [*Halls*, 1976; *Kirschvink*, 1980] and Bingham statistics [*Onstott*, 1980]. For two of these sites, planes and lines were combined by transforming the line data into planes by method 2 of *Kirschvink* [1980]; the combinations passed the F-test at the 95% confidence level. The remaining four sites were entirely planar data.

Twenty-six of the original 34 sites yield reliable site mean directions. All the sites have normal polarity, consistent with their Late Cretaceous age [e.g., *Harland et al.*, 1982]. Eight sites were excluded from the final statistical analysis for the following reasons. Four sites, including three sites from volcanic breccias and conglomerates, exceeded an arbitrary $\alpha 95$ limit of 15° (Table 1). Three sites have virtual geomagnetic poles (VGP) which are removed from the mean by more than 38°; this is approximately twice the angular dispersion

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TABLE 1. Paleomagnetic Data for all Sites

Site	Rock	λ	ϕ	S/N	B_m/T_m	GDEC	GINC	SC	SDEC	SINC	α95	k	PLAT	PLONG
88AV01#	D	47.29	111.94	9/9	6.0/220	34.0	79.1	0/0	34.0	79.1	4.2/5.0	-106.9/-0.6	62.6	-86.0
88AV02+	D	47.13	111.88	8/8	5.5/275	42.0	38.2	160/38	6.0	66.4	4.96	125.65	85.6	-47.9
88AV03	F	47.13	111.88	7/7	46.0/590	45.6	27.1	157/38	25.7	60.2	5.15	138.41	70.7	-13.2
88AV04	D	47.22	111.77	8/8	9.0/200	330.3	73.6	0/0	330.3	73.6	7.18	60.46	68.7	-155.4
88AV05	F	47.13	111.86	8/8	45.0/525	3.4	47.4	70/28	32.9	70.3	12.54	20.46	68.2	-53.4
88AV06	F	47.13	111.91	8/8	15.0/300	4.8	25.8	114/47	335.3	66.7	3.55	244.64	73.5	174.7
88AV07	D	47.13	111.86	8/8	20.0/530	10.4	67.6	0/0	10.4	67.6	4.20	175.31	82.4	-51.8
88AV08*	F	47.13	111.86	6/8	47.5/575	24.3	65.7	70/28	100.3	70.6	5.50	149.57	31.9	-69.9
88AV09	F	47.16	111.82	7/7	12.5/450	319.7	55.0	30/13	329.5	66.9	3.10	381.14	69.7	176.1
88AV10	F	47.16	111.82	7/9	27.5/450	9.4	69.8	190/20	326.8	61.7	3.94	236.06	66.3	160.2
88AV11	F	47.17	111.82	8/8	23.0/525	333.6	32.2	37/42	8.5	64.5	7.25	59.33	84.1	-17.1
88AV12	F	47.14	111.86	8/8	45.0/570	341.5	57.3	80/27	309.6	82.9	4.41	158.96	54.7	-130.7
88AV13	L	47.47	111.83	8/8	3.5/300	356.5	74.5	0/0	356.5	74.5	6.02	85.52	76.4	-119.1
88AV14*	D	47.28	111.98	8/8	4.5/250	32.5	50.5	0/0	32.5	50.5	12.07	22.00	60.5	-0.8
88AV15*	VC	47.10	111.96	8/8	12.5/460	50.9	41.5	135/7	51.6	48.5	134.59	1.15	46.7	-16.8
88AV16	F	47.13	111.91	6/7	28.0/490	343.3	6.5	85/42	337.8	47.3	12.91	27.87	64.5	118.8
88AV17+#	D	47.13	111.91	10/10	5.5/220	13.3	26.1	114/47	353.3	71.0	5.0/9.5	-52.7/-2.6	80.7	-136.1
88AV18#	F	47.13	111.91	9/9	32.0/440	10.6	21.0	114/47	352.8	65.3	8.7/9.9	-27.3/-0.5	85.1	163.8
88AV19	L	47.48	111.85	8/8	7.0/370	333.1	78.2	0/0	333.1	78.2	2.20	632.60	65.9	-137.2
88AV20	D	47.23	111.76	6/8	6.5/ -	345.4	77.4	0/0	345.4	77.4	6.77	98.87	69.8	-129.1
88AV21#	L	47.42	111.74	6/8	6.5/350	277.4	77.5	0/0	277.4	77.5	2.3/3.0	-485.3/-1.0	45.1	-146.5
88AV22	D	47.18	111.81	8/8	3.5/140	4.8	68.3	0/0	4.8	68.3	9.46	35.24	84.7	-77.9
88AV23*	VB	47.16	111.83	9/9	18.0/420	294.2	50.5	20/30	306.1	80.2	67.30	1.55	55.5	-139.7
88AV24*	F	47.14	111.89	8/8	- /615	342.4	7.8	85/45	334.8	51.2	16.22	12.62	65.5	128.7
88AV25	F	47.17	111.82	7/8	30.0/520	33.7	37.6	170/46	344.2	54.9	10.05	37.02	73.3	118.9
88AV26	F	47.17	111.84	9/9	20.0/410	343.3	53.9	52/14	354.4	66.5	4.40	137.90	85.9	-173.9
88AV27+	D	47.13	111.87	7/7	7.5/ -	35.9	33.9	190/43	1.3	40.9	2.68	508.38	66.3	65.2
88AV28+	D	47.13	111.87	7/7	5.5/ -	51.8	48.3	190/43	349.6	58.1	3.56	289.30	78.7	114.1
88AV29*	VB	47.10	111.95	9/9	32.5/510	94.8	33.8	205/28	81.1	59.0	31.85	3.57	33.4	-46.6
88AV30*	F	47.14	111.85	10/10	30.0/600	55.4	64.0	350/25	66.1	40.2	7.02	48.35	32.6	-21.3
88AV31	F	47.16	111.82	8/8	57.5/570	350.7	49.7	35/25	24.3	61.8	3.73	221.72	72.4	-17.2
88AV32	F	47.16	111.82	7/8	56.0/570	320.4	51.0	25/25	350.2	70.8	3.96	233.26	79.9	-145.5
88AV33#	L	47.32	111.91	5/5	6.0/ -	323.3	63.6	0/0	323.3	63.6	4.3/4.9	-190.2/-0.5	64.8	167.0
88AV34#*	L	47.31	111.93	3/3	7.0/ -	16.0	8.3	0/0	16.0	8.3	5.1/6.0	-216.0/-0.7	44.7	45.3

+ Tilted dikes included with flows for fold test.

Sites for which Bingham statistics were used to analyze intersecting plane data.

* Sites not used in final paleopole analysis (see text).

Rock: F,flow; D,dike; L,laccolith; VC,volcanic conglomerate; VB,volcanic breccia. λ,ϕ: site latitude (°N) and longitude (°W). S/N: samples used/collected. B_m/T_m: average median destructive induction (B_m) and temperature (T_m) in mT/°C. GDEC,GINC: site mean declination and inclination before structural correction (geographic coordinates). SC: structural correction, strike/dip (dip direction is 90° clockwise from strike). SDEC,SINC: site mean declination and inclination after structural correction (stratigraphic coordinates). α95: Fisher (α95) and Bingham (α31/α32) 95% confidence limits. k: concentration parameter for Fisher (k) and Bingham (k1/k2) distributions. PLAT,PLONG: VGP latitude (positive north) and longitude (positive east).

inherent in paleosecular variation over the past 5 m.y. at the sampling paleolatitude [*McElhinny and Merrill*, 1975]. These three sites most likely acquired their magnetization during a reversal or anomalous behavior of the geomagnetic field and are not representative of normal paleosecular variation. Finally, one site was discarded because it did not yield stable characteristic remanent directions or demagnetization circles regardless of thermal or af demagnetization experiments. This last site also had an extreme susceptibility to acquiring viscous remanent magnetization during the cleaning process.

We measured acquisition of saturation isothermal remanent magnetization (IRMs) to study the magnetic mineralogies which carry NRM and used normalized NRM/IRMs coercivity spectra to investigate magnetic grain size distributions and domain structure [*Dunlop*, 1983; *Fuller et al.*, 1988; *Lowrie and Fuller*, 1971]. Generally, these techniques showed that the Adel rocks contain bimodal distributions of titanomagnetites, with many intrusives being dominated by multi-domain paleomagnetic behavior (see *Gunderson* [1989] for details).

PALEOMAGNETIC RESULTS

For discussion, the Adel Mountain volcanic field is conveniently divided into three geologic domains: flows, dikes and laccoliths, and volcanic conglomerates and breccias. The paleomagnetic data for all sites are summarized in Table 1.

Flows

Most of these 16 sites are in variously dipping blocks along the Missouri River (Figure 2). Specimens from these volcanic flows generally exhibit excellent magnetic stability during demagnetization. Nearly all specimens have single component magnetizations with maximum angular deviations commonly less than 3° over their entire coercivity and/or thermal unblocking spectra (Figure 3*a*). Secondary magnetic components were isolated from just four sites. All are probably present-day overprints which can be removed with inductions of 10–20 mT or temperatures less than 300°C

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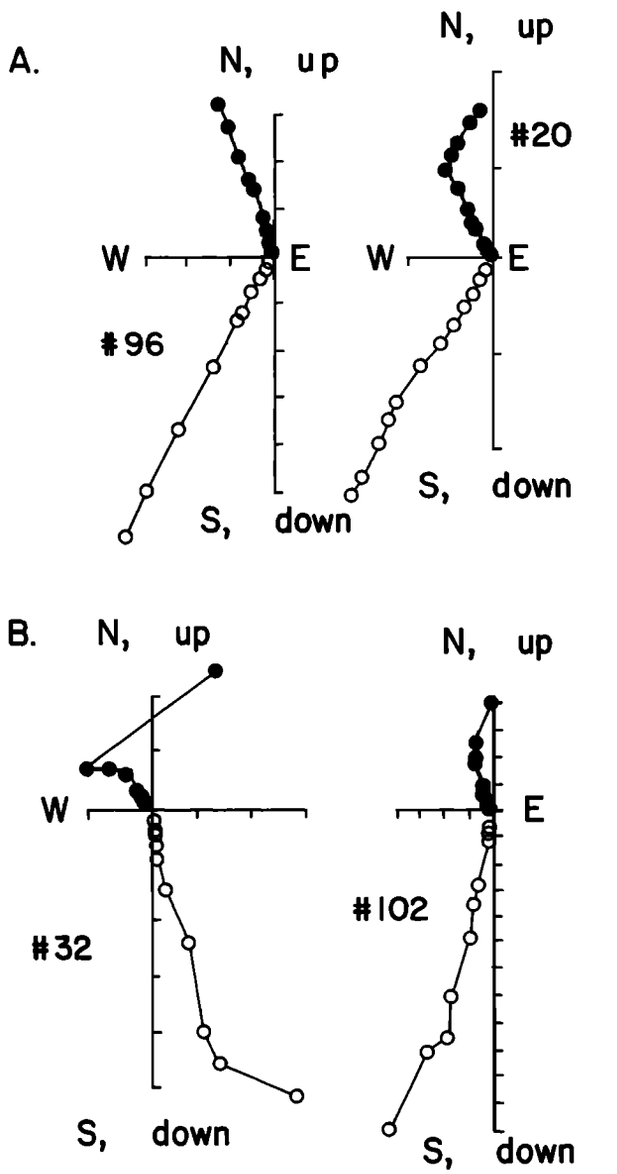


Fig. 3. Vector plots of representative samples. Inclinations (open circles), shown on the vertical section, are projected into the plane of declination (solid circles). (*a*) From flows, scale divisions of 1 A/m. Specimen 96 was af demagnetized from NRM to 100 mT. Specimen 20, thermal demagnetized from NRM to 600°C, loses secondary component at 350°C. (*b*) From dikes/laccoliths, scale divisions of 0.1 A/m. Specimen 32, af demagnetized from NRM to 25 mT, turns to the origin at 10 mT. Specimen 102 was thermally demagnetized from NRM to 600°C.

(Figure 3*a*). Average median destructive inductions and temperatures for these rocks are 35 mT and 540°C, respectively.

IRM acquisition curves indicate that titanomagnetites are the primary carrier of NRM, with saturating inductions of 0.5–0.8*T* [*Gunderson*, 1989]. However, coercivities exceeding 100 mT and high unblocking temperatures (greater than 600°C) for about 80% of the specimens demonstrate that 10–30% of the NRM resides in ilmenohematites. Where present and separable, the paleomagnetic direction of ilmenohematite invariably parallels that of titanomagnetite.

All of the flows have measurable dip. Because four dikes (sites 88AV02, 88AV17, 88AV27, and 88AV28) within the volcanic pile have obviously been tilted along with the extrusives, we include them with the reliable flows for a fold test. Thus, structural corrections were applied to all flows and the four tilted dikes. The flows generally have southerly dips ranging from 10° to 45° with an average of about 30°. Figure 4, which includes 13 reliable VGPs from the flows and four from the tilted dikes, compares the site distributions and their means before and after tilt correction. The structural correction appears very convincing (Figure 4). Excluding the three flows that do not pass the final reliability criteria, the mean direction for the remaining 13 flows and four tilted dikes is $D = 6.6°$, $I = 43.6°$, $\alpha 95 = 12.29°$, $k = 9.40$. After structural correction the mean direction is $D = 354.4°$, $I = 64.6°$, $\alpha 95 = 5.77°$, $k = 39.26$. Analysis by *McElhinny's* [1964] method indicates that the fold test is statistically significant at the 99% confidence level. Thus the characteristic remanent magnetization is older than the tilting and most likely a primary thermal remanent magnetization (TRM).

The mean direction from the 13 flows is $D = 353.3°$, $I = 66.3°$, $\alpha 95 = 6.61°$. Averaging their structurally corrected VGPs gives a mean pole at 84.8°N, 190.3°E, $\alpha 95 = 10.01°$. Table 2 presents these and the following statistics in a convenient form.

Dikes and Laccoliths

The unweathered dikes and laccoliths they feed (Figure 2) represent the latest stage of igneous activity in the Adel Mountain volcanic field. Our K-Ar date on biotite from one of these shonkinite dikes that crosscuts the volcanic field is 71.2 ± 2.7 Ma. Many dikes are mappable for several kilometers; almost all the dikes, except in the center of the volcanic field, are vertical. The laccoliths, distributed over more than 200 km^2, have horizontal contacts with surrounding sediments and vertical columnar jointing. Thus, our sites in these rocks do not appear to be disturbed by the tilting and slight folding which affect the flows in the center of the volcanic field. Our geological observations indicate that these sites require no structural corrections.

NRM intensities of specimens from dikes (including the four deformed dikes grouped with the flows) and laccoliths are very similar to those of the flows, but their af coercivity spectra are much lower. Their median destructive inductions are about 4–10 mT. Figure 3*b* presents representative demagnetization diagrams for these rocks (compare with Figure 3*a*). The dikes and laccoliths provide consistent within-site paleomagnetic directions; specimens generally have maximum angular deviations less than 6° during the last several demagnetization steps. Only one site had a consistent secondary overprint. The dikes and laccoliths have saturating inductions ranging from 0.1 to 0.2*T*, which indicates titanomagnetites are the dominant NRM carrier. Magnetizations in these rocks were completely unblocked by 560°C (Figure 3*b*) during thermal cleaning.

The mean direction for 13 reliable sites (four are from the center of the volcanic field and are tilt corrected) in these late dikes and laccoliths is $D = 351.1°$, $I = 70.5°$, $\alpha 95 = 6.86°$. The mean direction for the dikes and laccoliths is only 4.3° from the mean of the flows, and by the method of *McFadden and Lowes* [1981] the two distributions share a common mean at the 95% confidence level (Figure 5). Averaging VGPs of these 13 sites yields a mean pole at 79.1°N, 219.2°E, $\alpha 95 = 9.95°$ (Table 2).

Because of their lower median destructive inductions and the lack of a direct fold test, some may question whether the dikes and laccoliths yield primary magnetizations or present-day viscous remanent magnetizations. However, NRM/IRMs ratios are all about 10^{-2} [*Gunderson*, 1989], consistent with a thermal origin for the NRM [*Dunlop et al.*, 1984; *Fuller et al.*,

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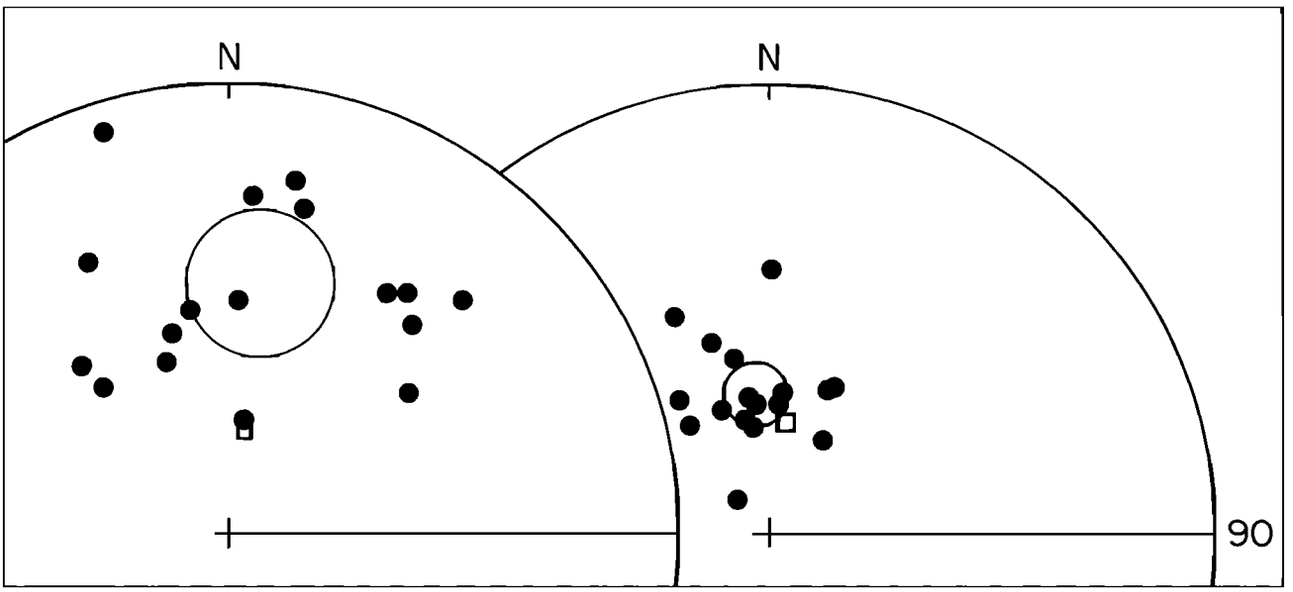


Fig. 4. Stereonets showing distribution of 17 site means from 13 flows and four dikes before the tilt correction (on the left) and after tilt correction (on the right).

TABLE 2. Tilt-Corrected Paleomagnetic Data From the Adel Mountain Volcanic Field

	N	DEC	INC	*k*	α95	s63	PLAT	PLONG	dp	dm
Flows	13									
(Dir)		353.0	66.3	40.3	6.61	12.75	85.1	178.5	8.9	10.8
(VGP)			18.1	10.01	19.07	84.8	190.3			
Dikes	13									
(Dir)		351.1	70.5	37.5	6.86	13.22	80.6	214.9	10.3	11.9
(VGP)			18.3	9.95	18.95	79.1	219.2			
Combined	26									
(Dir)		352.2	68.4	39.4	4.58	12.91	83.4	200.9	6.5	7.7
(VGP)			18.4	6.80	18.93	82.2	209.9			

N is the number of sites in a mean, Dec and Inc are the site-mean declination and inclination, *k* is the concentration parameter for Fisher distributions, α95 is the semiangle of the 95% confidence cone in degrees, s63 is the angular standard deviation in degrees, PLAT and PLONG are the latitude (positive north) and longitude (positive east) of the virtual geomagnetic pole, dp is the error in the VGP colatitude, dm is the error perpendicular to dp. (Dir) and (VGP) indicate that the associated statistics [*Fisher*, 1953] are calculated from directions and virtual geomagnetic poles, respectively.

1988]. Moreover, at the 95% confidence level, the mean direction for these sites does not overlap a magnetic pole centered on the Earth's rotation axis (Figure 5). Convincing evidence for the magnetic stability of the dikes and laccoliths is also offered by the tilted dikes included in the discussion of the volcanic flows and the fold test. The tilted dikes are identical in composition, appearance, af-spectra, and unblocking temperatures to the undeformed dikes and laccoliths. Before tilt correction the deformed dikes have site-mean directions which overlap neither the present day field or the rotation axis. They are within the group of rocks that pass the fold test at the 99% confidence level and retain stable magnetic components which are clearly not present-day. The geologic and paleomagnetic evidence strongly indicate that the characteristic remanent directions isolated from dikes/laccoliths are primary magnetizations.

Conglomerates and Breccias

To test their paleomagnetic reliability and to perform a classic conglomerate test [*Graham*, 1949], we collected three sites (88AV15, 88AV23, and 88AV29) in the many volcanic conglomerates and breccias of the Adel Mountain volcanic field. The matrix of the conglomerates and breccias is very pebbly. Specimens collected from the matrix of these rocks demonstrate large anisotropies in their magnetizations. One section of a specimen from the matrix often has a different direction of magnetization than an adjacent section of the specimen. Even so, they often maintain surprisingly stable directions; this is likely due to simultaneous cleaning of randomly oriented stable components within the pebbly matrix. Specimens from clasts of volcanic rock within the conglomerates and breccias have the very stable

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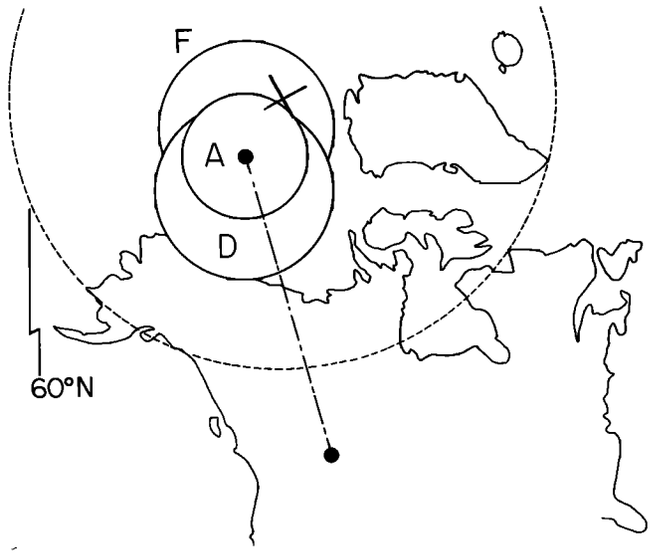


Fig. 5. Stereographic projection showing the mean VGPs and their 95% confidence limits for F and D, the flows and dikes/laccoliths, respectively; A, the pole from combining sites in the flows and dikes/laccoliths. The dashed line connects the Adel Mountain sampling area with the mean Adel pole; the cross shows the spin axis.

paleomagnetic behavior common to sites collected in the volcanic flows. Clasts have very stable magnetizations but no directional similarity between individual clasts in a conglomerate or breccia. All three sites yield mean directions with $\alpha 95$ greater than 30° (Table 1), providing qualitatively positive results for the conglomerate test. Individual clasts obviously acquired and retain a predepositional primary magnetization. Although not included in further statistical analysis, these sites provide additional strong support that the Adel Volcanic rocks carry reliable primary magnetic directions.

DISCUSSION AND CONCLUSIONS

Adel Mountain Volcanic Field Paleomagnetic Pole

The pole positions estimated from our two geologic domains (flows and dikes combined with laccoliths), compared by using the conditional test of *McFadden and Lowes* [1981], are not statistically different at the 95% confidence level. However, the dikes and laccoliths clearly crosscut the flows and yield a younger K-Ar date (71.2 ± 2.7 Ma versus 81.1 ± 3.5 Ma). Overall, the dikes and laccoliths are younger than the flows. The paleopole estimated from the dikes and laccoliths (79.1°N, 219.2°E, $\alpha 95$ = 9.95°, n = 13) is about 10° from *Globerman and Irving's* [1988] Cretaceous reference pole. Yet, the paleopole estimated from the older volcanic flows (84.8°N, 190.3°E, $\alpha 95$ = 10.01°, n = 13) is closer to *Diehl et al.'s* [1983] Paleocene reference pole (81.5°N, 192.6°E, $\alpha 95$ = 3.2°). Thus the older paleopole is displaced toward the Tertiary segment of the apparent polar wander path. This apparent departure of the APW path from smooth behavior between well defined cusps could be very interesting [e.g., *Gordon et al.*, 1984]. Accepting it requires that both sets of VGPs (flows and dikes) sufficiently averaged paleosecular variation to a constant representative value for the Late Cretaceous. However, the flows only yield 13 VGPs, the dikes and laccoliths provide 13 additional VGPs. Most likely, neither set yields a sufficient number of VGP to adequately average paleosecular variation. Such a situation is not unexpected; many apparently episodic volcanic fields fail to randomly sample the magnetic field during their duration whether measured by radiometric or stratigraphic methods. Limited subsets fromextensive volcanic fields may not sufficiently sample paleosecular variation.

Two modern detailed and comprehensive studies of volcanic rocks indicate how difficult it can be to determine an accurate paleomagnetic pole. First, *Diehl et al.* [1988] present 239 mid-Tertiary VGPs from 16 different studies, yet the reverse and normal polarity subsets yield paleomagnetic poles that are different at the 95% confidence level. Second, *Calderone et al.* [1990] estimate that they obtained only 65 reliable VGPs from 190 middle Miocene lava flows exposed in the Mojave-Sonora desert of Arizona and California. Limited subsets of either of these two studies would not always yield an accurate pole [cf. *Calderone and Butler*, 1984]. In addition, the Oligocene Hog Heaven volcanic field of northwestern Montana yields only 12 reliable VGPs from 25 sites [*Sheriff*, 1989] and fails to provide a reliable estimate of the Oligocene rotation axis. Thus recent studies prove it may be difficult to make accurate estimates of the ancient spin axis from reasonably extensive volcanic fields.

Although our radiometric dates indicate a difference of as much as 10 m.y. between the early flows and late dikes, we choose to combine all of our reliable VGPs and calculate a mean paleomagnetic pole for the whole Adel Mountain volcanic field. Combining the VGPs increases the likelihood that the pole position contains an adequate sampling of paleosecular variation.

Combined, the 26 reliable site VGPs (Figure 6) give a single paleomagnetic pole position at 82.2°N, 209.9°E, $\alpha 95$ = 6.80°, k = 18.38 (Table 2). The angular dispersion (circular standard deviation) of the 26 VGPs is 18.9° with lower and upper 95% confidence limits of 15.8° and 23.3° respectively [*Cox*, 1969]. This is very similar to the 18.6 ± 1.1° of dispersion predicted by paleosecular variation model G [*McFadden et al.*, 1988] for the paleolatitude of the Adel Mountain volcanic field. All our sites are of normal polarity which is probably not an indication of poor sampling of the ancient magnetic field because the field maintained normal polarity for about 80% of the time between 71 Ma and 81 Ma [*Harland et al.*, 1982]. Our sampling scheme of collecting sites across structure, good stratigraphic separation of sites in volcanic flows, and data that yield the expected angular dispersion add confidence that the Adel Mountain volcanic rocks sufficiently sampled paleosecular variation.

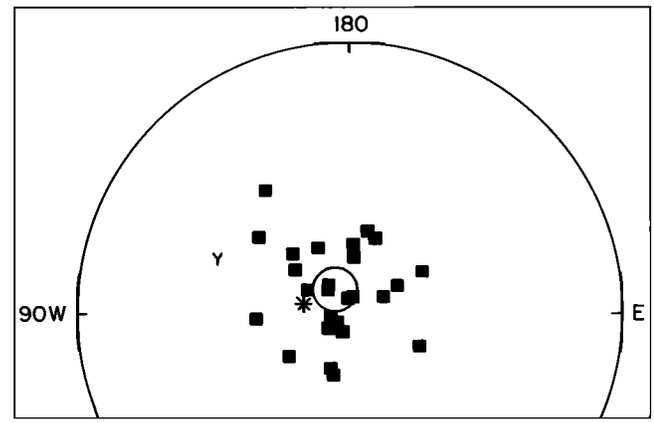


Fig. 6. Stereographic projection centered on the rotation axis showing the distribution of 26 reliable VGPs and their associated circle of 95% confidence. The asterisk marks the present-day magnetic pole; the "Y" marks the sampling locality.

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Figure 7 illustrates the new paleopole for the Adel Mountain volcanic field (82.2°N, 209.9°E, $\alpha 95$ = 6.80°), the mid-Cretaceous reference pole [*Globerman and Irving*, 1988], and the Early and mid-Tertiary reference poles [*Diehl et al.*, 1983, 1988]. The Late Cretaceous (81–71 Ma) Adel Mountain volcanic field paleopole lies 2.5° from the Paleocene (55–67 Ma) reference pole and 11.6° from the mid-Cretaceous (85–136 Ma) pole.

Comparison With Other Late Cretaceous Pole Positions

The youngest pole positions included in *Globerman and Irving's* [1988] preferred mid-Cretaceous reference pole are from the 88–100 Ma alkalic intrusives of Arkansas and the 85–89 Ma Niobrara Formation of Wyoming, Colorado, and Kansas [*Shive and Frerichs*, 1974]. Four estimates of the Late Cretaceous pole position for North America have been used in recent compilations; all are candidates for inclusion with the Adel Mountain pole in calculating a reference pole for the Late Cretaceous. These four are (1) the Elkhorn Mountains volcanic field (75–80 Ma) of west central Montana [*Hanna*, 1967, 1973], (2) the Livingston Formation (~80 Ma) pole from southwestern Montana [*Swenson and McWilliams*, 1989], (3) the Maudlow Formation (75–83 Ma) pole from southwestern Montana [*Swenson and McWilliams*, 1989], and (4) the ~72 Ma pole from the Roskruge volcanic rocks of southern Arizona [*Vugteveen et al.*, 1981]. None of these four are choice candidates for increasing the accuracy of the Late Cretaceous reference pole.

Hanna's [1967] data from the Cretaceous (~78 Ma) Elkhorn Mountains volcanic field of southwestern Montana are concordant with *Globerman and Irving's* [1988] preferred mid-Cretaceous pole. However, *Hanna's* [1967, 1973] calculations are not correct. Although he presents a paleopole (69°N, 171°W) which is concordant with other Cretaceous poles, it cannot be obtained from the site means and structural corrections presented [*Hanna*, 1965, 1967, 1973]. Because of this problem, the Elkhorn Mountains volcanic field pole has not been included in recent compilations of Cretaceous pole positions. Also, the Elkhorn Mountains volcanic rocks lie in the Helena salient [e.g., *Harrison et al.*, 1974] west of the leading edge of east-directed thrust belt deformation. These rocks are carried in thrust plates and have likely experienced rotations about nearby vertical axes during Late Cretaceous to Tertiary thrusting. *Eldredge and Van der Voo* [1988] show that thrust sheet rotations due to buttressing against the Rocky Mountain Foreland are common in the Helena salient of the fold and thrust belt. This region has also undergone ~30% crustal extension since the Early Tertiary [e.g., *Sheriff et al.*, 1984; *Lageson and Sheriff*, 1986]. Although new results [*Diehl et al.*, 1989] provide precise data from the Elkhorn Mountains volcanic field, the structural complexity and high likelihood of rotations undetectable by traditional geologic techniques render the Elkhorn Mountains volcanic rocks unsuitable for inclusion in a Late Cretaceous reference pole.

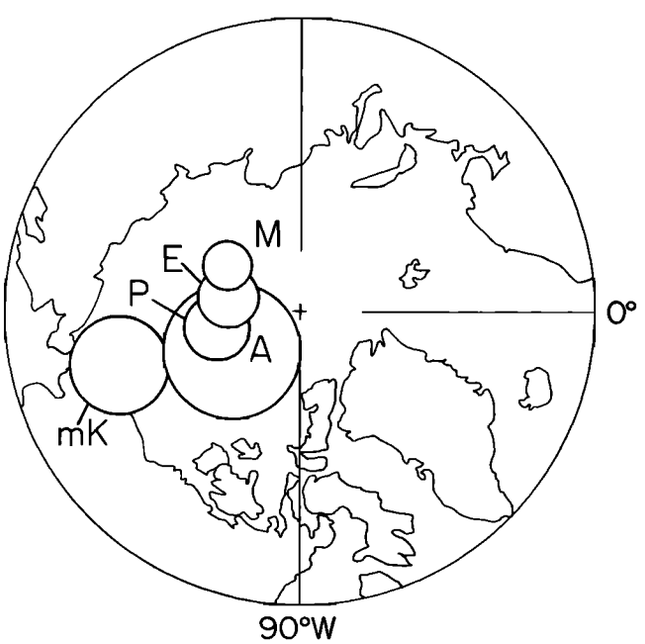


Fig. 7. Stereographic projection showing reference poles: mK, mid-Cretaceous; P, Paleocene; E, Eocene; M, Oligocene-Miocene, and A, the new Late Cretaceous pole from the Adel Mountain volcanic field. Circles represent 95% confidence limits.

The Livingston Formation and Maudlow Formation poles are provided by *Swenson and McWilliams* [1989]. The Livingston Formation yields a high coercivity magnetization which fails the fold test [*Swenson and McWilliams*, 1989]. This overprint direction may be either Cretaceous or Paleocene in age and has no good paleohorizontal control [*Swenson and McWilliams*, 1989]. Thus it is not suitable for inclusion in a reference pole. The Maudlow Formation yields high precision paleomagnetic results. However, as noted by *Swenson and McWilliams* [1989], there is a high probability that the sampling area (46.1°N, 248.9°E) has been rotated around a nearby vertical axis. The sampling area, like the Elkhorn Mountains volcanic field, lies within the Helena salient of the fold and thrust belt. Most likely, the Maudlow basin lies above a duplex system that is rooted in a major detachment surface [see *Schmidt et al.*, 1988]. The minimum displacement of the Maudlow basin during Late Cretaceous to Eocene thrusting is probably 20–30 km and very likely much more (*D. Lageson*, personal communication, 1989). Also, the paleomagnetic sites in the Maudlow Formation are within 20 km of documented rotations of Cretaceous rocks in the Helena salient caused by thrusts buttressing against the Rocky Mountain foreland [*Eldredge and Van der Voo*, 1988]. As *Swenson and McWilliams* [1989] point out, the Maudlow Formation has a declination near the expected direction. This is a good test for discerning rotation of the sampling area, but we believe the possible structural problems and likely major transport of these rocks makes them poor candidates for inclusion in a Cretaceous reference pole.

Vugteveen et al. [1981] provide another Late Cretaceous paleopole at 73.6°N and 176.0°E (dp = 6.2°, dm = 8.8°, n = 9). This pole is from the ~72 Ma Roskruge Mountains volcanic rocks of southern Arizona. The Roskruge pole is about 7° from both the mid-Cretaceous and Paleocene reference poles; its 95% confidence limits overlap both (Figure 1). The Roskruge Mountain volcanic rocks provide modern high-quality data with good precision, but the rocks suffer from structural complications similar to rocks from the Elkhorn Mountains and Maudlow basin. Noting that they are uncertain whether any rotation has accompanied the ~20% extension which displaces the Roskruge volcanic rocks from the craton, *Vugteveen et al.* [1981] suggest their result be regarded as preliminary. *Vugteveen et al.* [1981] also present results from the nearby (~100 km) Paleocene Gringo Gulch volcanic rocks which are consistent with the Paleocene reference direction. Thus it is likely that the Roskruge volcanic rocks did not experience significant rotation about a nearby vertical axis during their Tertiary translation. Like the Maudlow basin case, this is a good test for local tectonic rotation but the structural uncertainty resulting from the Tertiary transport of these rocks reduces their value for inclusion in a Cretaceous reference pole.

For the reasons outlined above, we believe that the

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paleopoles for the Late Cretaceous of North America from the Elkhorn Mountains volcanic rocks [*Hanna*, 1965, 1967, 1973], the Livingston and Maudlow Formations [*Swenson and McWilliams*, 1989], and the Roskruge Mountains volcanic rocks [*Vugteveen et al.*, 1981] are probably not sufficiently accurate to include in a Late Cretaceous reference pole position. These examples were not chosen to discredit previous workers or attack anyone's analytical techniques. Rather, they are the poles that have been used in recent compilations for the Late Cretaceous of North America. Our examples emphasize the importance of evaluating, maintaining and improving the reference poles used to construct apparent polar wander paths. In general, paleomagnetic poles from upper plate (hanging wall) rocks that have undergone major crustal-scale deformation are probably not good candidates for inclusion in paleomagnetic reference poles. We recognize that paleomagnetic results from a single volcanic field, such as the Adel Mountain volcanic field, are also less than ideal for the required accuracy of a reference pole. However, until more results from Late Cretaceous rocks are available, the Adel Mountain pole is most likely the best estimate of the Late Cretaceous pole position for North America.

Comparison With Mid-Cretaceous Reference Pole

The age of the Adel Mountain paleomagnetic pole is about 76 Ma. The youngest paleomagnetic poles included in the mid-Cretaceous stillstand pole [*Globerman and Irving*, 1988] are from the 88–100 Ma alkalic intrusives of Arkansas and the 85–89 Ma Niobrara Formation of Wyoming, Colorado, and Kansas [*Shive and Frerichs*, 1974]. About 12° separates the mid-Cretaceous pole and the Late Cretaceous, Adel Mountain pole. The oldest contributors to *Diehl et al.'s* [1983] Paleocene reference pole are from the 67–62 Ma Moccasin Mountains and Judith Mountains of north central Montana. Less than 3° separates the Adel Mountain pole from the Paleocene pole. Therefore, about 12° of apparent motion of North America may have occurred in less than 9 m.y. between 76 Ma and 85 Ma. The period since about 76 Ma has been a time of limited apparent polar motion for North America. The Adel Mountain pole indicates that the major portion of rapid APW separating the J2-K track [*May and Butler*, 1986] from any Tertiary track occurred before the latest Cretaceous and in roughly half the time allowed in previous estimates. This period of rapid apparent polar wander, followed by little apparent polar motion in the Tertiary, was also preceded by a time of limited apparent motion. The paleopoles in the mid-Cretaceous reference pole [*Globerman and Irving*, 1988] are all quite close and indicate little apparent polar motion between about 88 Ma and 136 Ma. The cusp between the J2-K track and the probable Tertiary track is constrained by the Late Cretaceous (~76 Ma) Adel Mountain paleomagnetic pole. The cusp is older than 76 Ma and corresponds well with the timing of major plate reorganization ~80 Ma [*Coney*, 1978], increased absolute motion of the North American plate [*Engebretson et al.*, 1984], and the beginning of Laramide compressional deformation in the Rocky Mountains.

APPENDIX: RADIOMETRIC DATA

The following is a description of, and the analytical details from, our two new radiometric dates from the Adel Mountain volcanic field.

1. 88AV01 (Geochron B-8578) K-Ar (biotite). Shonkinite dike with phenocrysts of plagioclase, clinopyroxene, very fresh biotite and magnetite; 47.29°N, 111.94°W. This outcrop is obvious where the dike crosses the St. Peter Mission road in the NW 1/4, NW 1/4, sec. 25, T.18 N., R.3 W., Cascade Co., MT. Analytical data: K = 5.469% (^{40}K = 6.524 ppm); ^{40}Ar = 0.02754 ppm; ^{40}Ar/(total ^{40}Ar) = 43.5%. Dated by Geochron Laboratories, Inc. (biotite) 71.2 ± 2.7 Ma.

2. 88AV12 (Geochron R-8585) K-Ar (whole rock). This fresh shonkinite flow with no phenocrysts is at 47.14°N, 111.86°W, NW 1/4, NE 1/4, sec. 21, T.16 N., R. 2 W., Cascade Co., MT. Analytical data: K = 1.231% (^{40}K =1.468 ppm); ^{40}Ar = 0.007076 ppm; ^{40}Ar/(total ^{40}Ar) = 70.0%. Dated by Geochron Laboratories, Inc. (whole rock) 81.1 ± 3.5 Ma.

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